

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

Paper 1 Multiple Choice

9744/01

September 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **16** printed pages and **no** blank page.

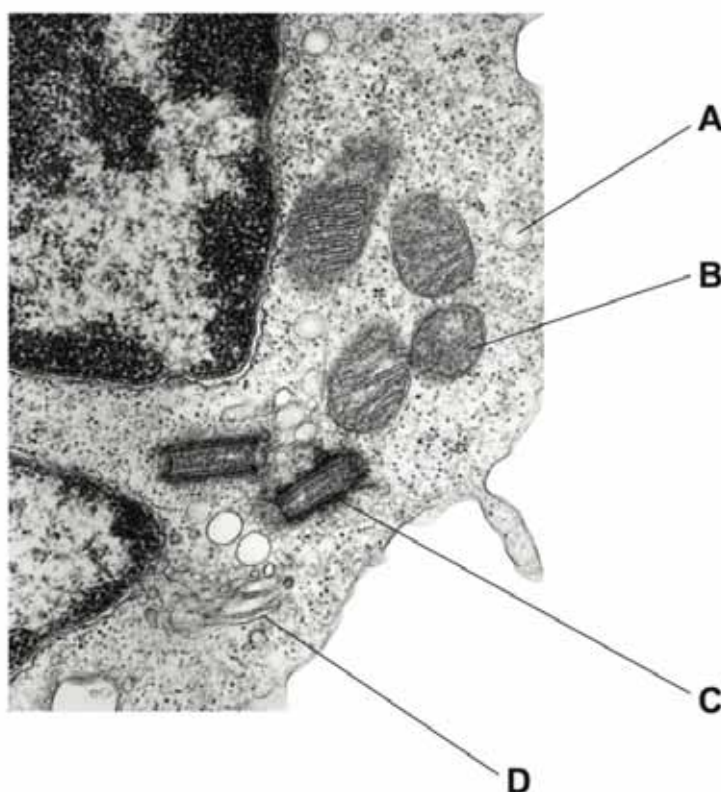
[Turn over

- 1 A student has drawn a cell structure as seen using a light microscope.
The magnification of the drawing is $\times 600$.
The length of the structure on the drawing is 6mm.

What is the actual length of the cell structure?

- A $1 \times 10^{-1} \mu\text{m}$ B $1 \times 10^0 \mu\text{m}$ C $1 \times 10^1 \mu\text{m}$ D $1 \times 10^2 \mu\text{m}$

- 2 The electron micrograph shows part of a eukaryotic cell.
Which of the labelled organelles is a site of protein synthesis?



- 3 The table shows some information about four carbohydrate polymers.

polymer	α -1,4 glycosidic bonds	α -1,6 glycosidic bonds	shape of molecule
1	✓	x	helical
2	x	✓	branched
3	✓	✓	helical
4	✓	✓	branched

key
✓ = present
x = absent

Which two polymers form starch?

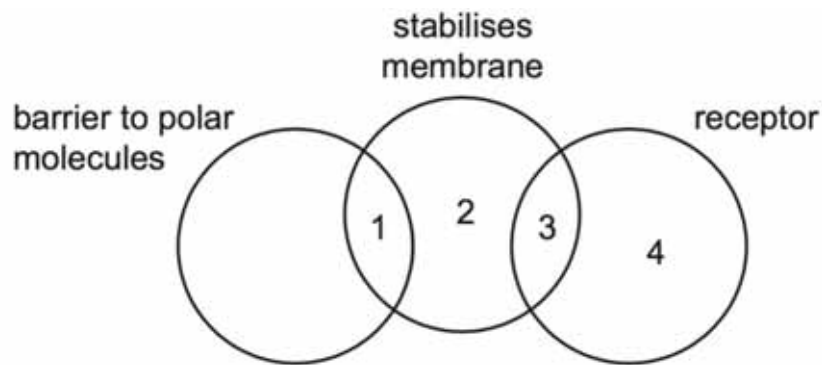
- A 1 and 2 B 1 and 4 C 2 and 3 D 3 and 4

- 4 When proteins are mixed with some organic solvents, hydrophobic interactions and hydrogen bonding are changed in the protein molecules.

Which levels of protein structure would be affected?

	level of protein structure			
	secondary	tertiary	quaternary	
A	✓	✓	x	key ✓ = affected x = not affected
B	✓	x	✓	
C	x	✓	✓	
D	✓	✓	✓	

- 5 Which row correctly links molecules in the cell surface membrane with their roles?



	1	2	3	4
A	glycolipid	cholesterol	glycoprotein	phospholipid
B	glycolipid	glycoprotein	phospholipid	cholesterol
C	glycoprotein	phospholipid	cholesterol	glycolipid
D	phospholipid	cholesterol	glycolipid	glycoprotein

- 6 The table below shows additional information about the enzymes that catalyse some of the reactions in respiration.

enzyme	information
fructose 1,6-bisphosphate aldolase	- four identical subunits - changes to any one of the subunits means that the enzyme cannot function
hexokinase	- one subunit - active site changes shape to enclose the reactants
phosphofructokinase	- four identical subunits - has allosteric sites in addition to an active site
phosphoglucose isomerase	- two identical subunits - has a cytokine function when secreted into the external medium
pyruvate kinase	- four identical subunits - ATP acts as an inhibitor to regulate glycolysis
triosephosphate isomerase	- two identical subunits - each subunit has 14 alpha helices and 8 beta-pleated sheets

A student made the following deductions using the information provided in the table:

- Phosphoglucose isomerase, when secreted, can have a non-catalytic role.
- Only three of the six enzymes display quaternary protein structure.
- The active site of phosphofructokinase will change shape to allow the enzyme to act as a regulator in glycolysis.
- Each enzyme is coded for by one gene.
- The reaction catalysed by hexokinase is an induced-fit mechanism.

How many of the student's deductions are correct and can be supported using the information provided?

- A 1 B 2 C 3 D 4

- 7 Polypeptide synthesis is based on sequences of three nucleotides, each specific for an amino acid.

Which row shows the correct nucleotide sequences for an amino acid?

	nucleotide sequence of		
	non-transcribed DNA strand	mRNA codon	tRNA anticodon
A	GGT	CCA	GGU
B	GGG	CCC	CCC
C	CCG	CCG	GGC
D	CCT	CCU	CCU

- 8 DNA is said to replicate in a semi-conservative way.

Results of Meselson and Stahl's experiments gave overwhelming support to this theory. They used *E. coli* which has a generation time of 20 minutes.

Here are the steps in their experiment but they are in the wrong order.

- P** All bacteria contain ^{15}N DNA.
- Q** All bacteria contain hybrid DNA (^{15}N DNA and ^{14}N DNA).
- R** Bacteria contain either all ^{14}N DNA or hybrid DNA.
- S** Bacteria grown in a ^{15}N medium for many generations.
- T** Bacteria transferred to a ^{14}N medium and sampled every 20 minutes.

Which sequence of letters shows the correct order of the steps in the experiment?

- A** P→Q→R→S→T
- B** P→S→T→R→Q
- C** S→P→T→Q→R
- D** S→P→T→R→Q

- 9 The table shows the mode of action of two antibacterial drugs that can affect the synthesis of proteins.

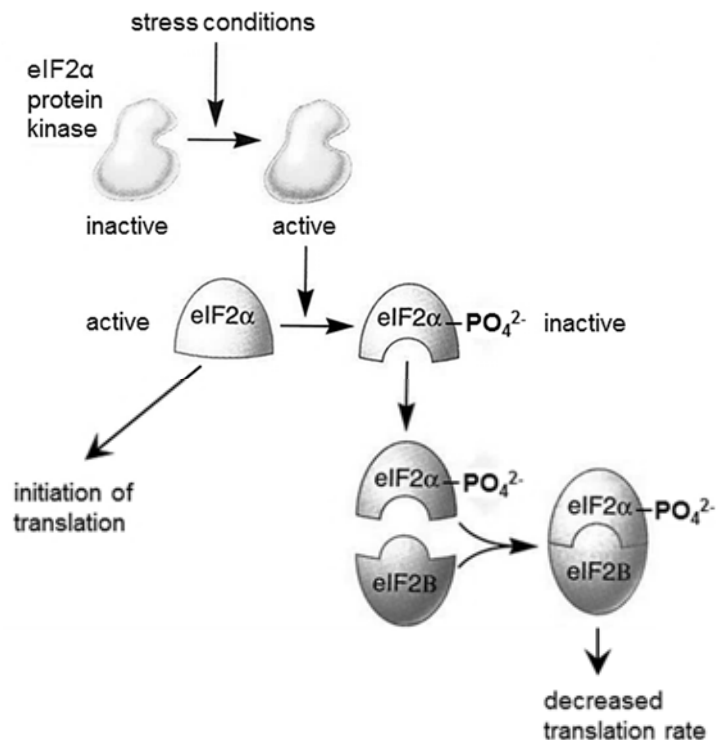
antibacterial drug	rifampicin	streptomycin
mode of action	binds to the RNA polymerase	causes errors in translation

If bacteria are treated with both drugs, what will be the immediate effects?

- 1 Transcription will stop, but faulty proteins may continue to be synthesised.
- 2 If translation has started, proteins may be faulty.
- 3 Translation will be inhibited.

A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

- 10 The diagram shows a mechanism by which gene expression is controlled during translation.



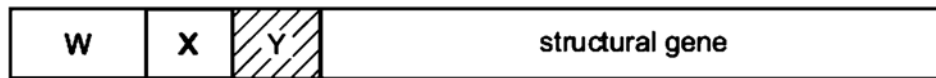
Which statements are correct?

- 1 Phosphorylation by eIF2α protein kinase changes the conformation of and activates eIF2α.
- 2 The concentration of eIF2B affects the rate of translation by inhibiting translation initiation.
- 3 This regulation results in decreased rate of mRNA translation under stress conditions.
- 4 The eIF2α-eIF2B complex is recognised by proteasomes for selective degradation due to the presence of the phosphate group.

A 1 and 2 B 1 and 4 C 2 and 3 D 3 and 4

11 The diagram represents a length of DNA from a prokaryote that includes a structural gene.

Parts of the length of DNA are labelled **W**, **X** and **Y**. They have different functions in the control of transcription of the structural gene.



What identifies the functions of parts **W**, **X** and **Y**?

	W	X	Y
A	operator	regulator	promoter
B	promoter	regulator	operator
C	regulator	promoter	operator
D	promoter	operator	promoter

12 Which of the following statements concerning lac operon is true?

- 1 Transcription of lac operon takes place all the time.
- 2 There is one single mRNA transcribed from the lac operon.
- 3 There is one start and one stop codon in the mRNA of lac operon.
- 4 The repressor molecule binds to the operator to turn off lac operon.

- A** 4 only
- B** 1 and 3
- C** 2 and 4
- D** 2, 3 and 4

13 Some events that take place during generalised transduction are listed below.

- 1 Bacterial host DNA is fragmented.
- 2 Bacterial DNA may be packaged in a phage capsid.
- 3 Recombination between donor bacterial DNA and recipient bacterial DNA.
- 4 Phage infects a bacterial cell.
- 5 Phage DNA and proteins are made.

Which sequence of events is correct?

- A** 4 1 3 5 2
- B** 4 1 5 2 3
- C** 4 5 2 3 1
- D** 4 5 1 3 2

14 What are the conditions in a human cell just before the cell enters prophase?

	number of molecules of DNA in nucleus	spindle present	nuclear envelope present
A	46	yes	no
B	46	no	yes
C	92	yes	yes
D	92	no	yes

- 15** The following table shows the chromosome numbers in the hybrids formed between cabbage (*Brassica oleracea*) and radish (*Raphanus sativus*).

<i>type of cell</i>	<i>no. of chromosomes per cell</i>
parental cabbage	18
parental radish	18
parental gametes	9
F ₁ hybrids	18
F ₁ gametes	18
F ₂ hybrids	36
F ₂ gametes	18
F ₃ hybrids	36

During which of the following stages can the occurrence of non-disjunction explain the results?

- A** formation of the F₁ gametes
- B** formation of the F₂ gametes
- C** fusion of the parental gametes
- D** fusion of the F₁ gametes

- 16** The statements are about genes and proteins, involved in breast cancer.

- The main protein coded by *BRCA1* gene inhibits the growth of breast cancer cells.
- The protein coded by the *p53* gene suppresses tumours.

Which combination of genes is most likely to result in breast cancer?

	gene	
	<i>BRCA1</i>	<i>p53</i>
A	x	x
B	x	✓
C	✓	x
D	✓	✓

key
 ✓ = normal active gene
 x = mutated gene

17 The following table shows the mRNA codons for six different amino acids.

mRNA codons	amino acid
AAA AAG	lysine
AGA AGG CGG	arginine
GGU GGA GGC GGG	glycine
CCU CCA CCC CCG	proline
UGG	tryptophan
UAU UAC	tyrosine

The base sequence of mRNA coding for part of a polypeptide is shown below.

U A U A A G A G G C C U U G G
 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
 ↑
start reading

From the information provided, which of the predictions stated below is **not** true?

- A The insertion of a nucleotide between positions 3 and 4 is expected to result in a greater change in the amino acid sequence than an insertion between positions 12 and 13.
- B The deletion of a nucleotide at position 5 would result only in an alteration of the second amino acid in the chain.
- C The substitution of a different nucleotide at position 12 would produce no alteration in the amino acid chain.
- D The substitution of a different nucleotide at position 13 would result in the alteration of one amino acid.

18 Skin colour in onions is controlled by two pairs of alleles **Ss** and **Rr**, which segregate independently. The allele **S** is dominant and must be present to allow development of pigment in the skin. In its absence, the onion skin is white. Allele **R** is dominant and gives red colour; the recessive **r** gives yellow colour.

What will be the ratio of phenotypes in the offspring of a cross between plants of genotypes **SsRR** and **ssrr**?

- A all red
- B 1 red : 1 yellow
- C 1 red : 1 white
- D 1 white : 2 red : 1 yellow

19 Which of the following would cause phenotypic variation among organisms of the same genotype?

- A** continuous variation within the species
- B** different varieties of the same species
- C** exposure to different environments
- D** mutation

20 A tall, green-stemmed plant with the genotype **TTrr** was crossed to a short, red-stemmed plant with the genotype **ttRR**. The F1 plants were allowed to self-fertilise. A χ^2 test was carried out on the results obtained for the F2 generation.

Part of the table of values for χ^2 is shown.

degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The value of χ^2 in this investigation was 7.6.

What is the probability of this value of χ^2 and do the results fit the expected ratio?

	probability	results fit expected ratio
A	between 0.001 and 0.01	no
B	between 0.01 and 0.05	yes
C	between 0.05 and 0.1	yes
D	between 0.1 and 0.5	no

21 In a series of experiments, actively photosynthesizing plants were supplied with labelled reactants.

- 1** water containing ^{18}O isotope
- 2** carbon dioxide containing ^{17}O isotope
- 3** carbon dioxide containing ^{13}C isotope

Where in the chloroplast would the products of photosynthesis from these reactants be formed?

	^{18}O	^{17}O	^{13}C
A	stroma	stroma	thylakoids
B	stroma	thylakoids	stroma
C	thylakoids	stroma	stroma
D	thylakoids	stroma	thylakoids

22 In an experiment, four tubes were set up as shown in the table below.

tube	contents
1	Glucose + homogenized animal cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvate + homogenized animal cells

If all other conditions are kept constant, which of the following shows the amount of ATP produced in each tube in **increasing** order?

- A** 1 – 3 – 4 – 2
- B** 2 – 3 – 4 – 1
- C** 4 – 2 – 3 – 1
- D** 3 – 2 – 1 – 4

23 Which of the following correctly describes the function of second messenger in signal transduction pathway?

- A** They relay the signal from the outside to the inside of the cell.
- B** They serve as transcription factors to activate the transcription process.
- C** They amplify the message by directly phosphorylating the cascades of proteins.
- D** They relay the message from the inside of the membrane throughout the cytoplasm.

24 Darwin's view of the process of evolution to form new species (speciation) has been reinforced by more recent discoveries in genetics and cell biology.

In this view, which sequence of events is considered most likely to lead to speciation?

A	adaptation of population → competition and predation leading to natural selection → behavioural isolation → sympatric speciation
B	adaptation of population → competition and predation leading to natural selection → behavioural isolation → allopatric speciation
C	competition and predation leading to natural selection → geographical isolation → adaptation of isolated populations → sympatric speciation
D	competition and predation leading to natural selection → geographical isolation → adaptation of isolated populations → allopatric speciation

25 Natural selection acts

- A** directly on an individual's genetic make-up, thereby changing the survival probability of the individual.
- B** on individuals by changing their genes so they are better able to adapt to their environment.
- C** on the structures, physiologies and behaviours expressed by individuals in a population to change allele frequencies.
- D** on phenotypes of individuals so that they change to adapt to their environment and pass on these changes to their offspring.

- 26 The map shows the distribution (shaded area) of the lizards belonging to the family Iguanidae. Most species of iguana are found in America but a few species inhabit Madagascar and the islands of Fiji (arrows at the bottom centre and bottom right of map).



Two observations were made about the different species of iguana:

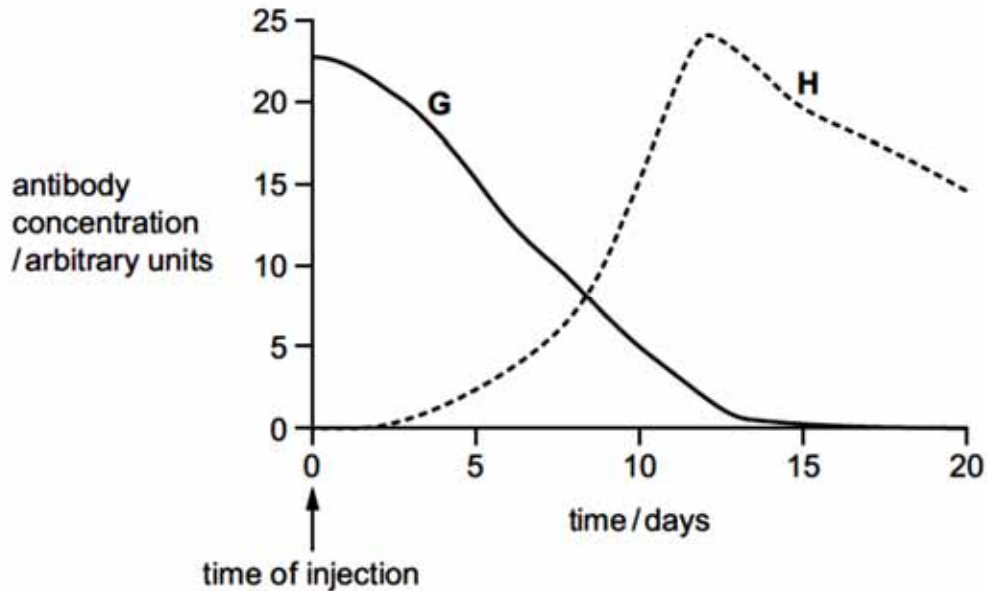
- 1 The various American iguana species shared more similar characteristics among themselves than with those iguana species on the island of Fiji.
- 2 The Madagascar iguana species was only distantly related to other lizard species on the African mainland.

Which observation and explanation best support the Darwinian concept of descent with modification?

	Observation	Explanation for the observation
A	1	The various American iguana species had a more recent common ancestor as compared to those iguana species on the island of Fiji that had diverged a longer time ago.
B	1	The various American iguana species shared more similarities among themselves as the degree of homology in their DNA was higher.
C	2	The Madagascan iguana species was reproductively isolated from the lizard species on the African mainland and thus diverged a long time ago.
D	2	The superficial similarities shared among the Madagascan iguana and the lizards on the African mainland were analogous, not homologous.

- 27 Two people, **G** and **H**, were each given an injection to protect them against a particular pathogen.

One person was injected with antibodies. The other person was injected with a vaccine. The graph shows the concentrations of the antibody against this pathogen in the blood of the two people, **G** and **H**, over a period of 20 days following the injection.



Which row correctly describes the type of immunity shown by **G** and **H**?

	G	H
A	artificial active immunity	artificial passive immunity
B	artificial passive immunity	artificial active immunity
C	natural active immunity	natural passive immunity
D	natural passive immunity	artificial active immunity

- 28 Which of the statements could describe **both** B-lymphocytes and T-lymphocytes?

- 1 They contain specific protein receptors in their cell surface membranes.
- 2 They differentiate into plasma cells.
- 3 They divide by mitosis.

- A** 1 and 3
B 1 only
C 2 and 3
D 2 only

- 29** What is the best definition of the greenhouse effect in the Earth's atmosphere?
- A** A naturally occurring effect by which shorter wavelength radiation is trapped
 - B** A naturally occurring effect by which longer wavelength radiation is trapped
 - C** An effect of pollution by which shorter wavelength radiation is trapped
 - D** An effect of pollution by which longer wavelength radiation is trapped
- 30** Global warming caused by the enhanced greenhouse effect is likely to have major consequences for arctic ecosystems. Which of the following are likely to occur in the arctic if the Earth's surface temperature rises?
- I** Decreased rates of decomposition of detritus
 - II** Increased range of predators from temperate regions
 - III** Increase in numbers of pest species and pathogens
- A** I and II only
 - B** I and III only
 - C** II and III only
 - D** I, II and III